

Alchemy system design – Wizard Yan’s Diary: The Remembrance Alchemy

System overview

Purpose

Alchemy is the primary knowledge-based system, through which the player learns to understand and manipulate elemental properties. Rather than collecting recipes, the player experiments with materials, observes outcomes, and gradually learns the principles governing their interactions. When designing it, I wanted to deconstruct the usual crafting trope of “collect materials, choose recipe, click craft”.

Core Alchemical Principle

Alchemy is a spatial reasoning system. Ingredients are charged with elemental energy, which moves the mixture through elemental space. The player learns to construct a trajectory toward the desired outcome.

Design goals

The rules should be simple and learnable. The system should not punish the player for failed experiments or require them to start over in case of a mistake or misclick. The outcomes should be deterministic. Each problem should have multiple valid solutions. Complexity should gradually increase throughout the game. The player should be provided with all the knowledge they need for a solution. The system is primarily a puzzle game.

Player mental model

The player should think of alchemy as navigating a space. When they choose ingredients, they pick movement patterns. When processing ingredients, they modify the movement patterns. They might try constructing a path from start to destination or the other way around. Depending on the topology they might need to avoid certain areas, which forces them to try a different combination of ingredients. Assuming that the length of a path is bounded, the player will have to search for a more efficient solution. If a path is not constructible with currently unlocked ingredients, they might need to search for new ingredients in other systems.

Model

Elemental space – Alchemical map

The game uses the five classical elements: fire, water, air, earth and Eather. They are color-coded: red for fire, blue for water, yellow for air, brown for earth and violet for Eather. The elemental space is a 3D coordinate system, in which the axes represent elemental energy: water – fire on horizontal axis and air – earth on the vertical axis. Eather energy corresponds to depth axis. A useful way to think about the elemental space is a set of 2D layers (maps), like floors of a skyscraper. At the beginning of an alchemy experiment, all elements are in balance – the point (0,0,0). Adding ingredients moves the mixture in three dimensions, which can be thought of as navigating a labyrinth. The number of ingredients per mixture is fixed but can be expanded further in the game (see Fuel section in this document).

Discovery zones are convex 2D subsets of layers. For sake of simplicity, they can be thought of as rectangular shapes covering some vertices. Landing the mixture in a discovery zone unlocks the corresponding discovery. These can be new ingredients, alchemical stations or their upgrades (i.e. capacity).

The game introduces elemental space first as 1D space (line), then adds more dimensions as the story progresses.

Water and Fire - First dimension

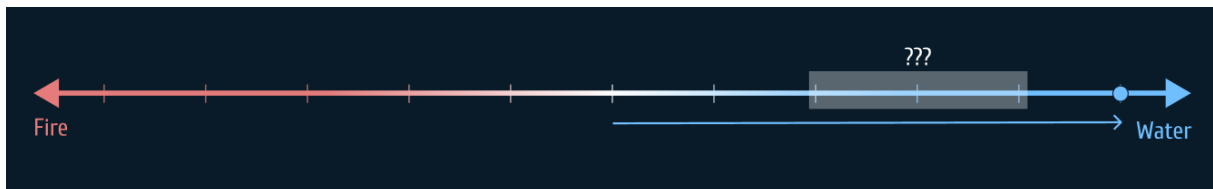
In the first dimension, the elemental space is a number line with water on the right end and fire on the left end. The experiment moves along a single axis, starting at 0. Ingredients push the mixture toward fire or water depending on their composition. 1 water energy corresponds to moving 1 unit right, 1 fire energy corresponds to moving 1 unit left. For an ingredient with composition (*water, fire*), the movement becomes (*water-fire*) units to the right. Discovery zones are intervals on number line, like [2,4]. When the mixture ends up with water energy between 2 and 4, the discovery is unlocked (see below for an example).

Assume the player has these ingredients at their disposal:

- **Water crystal** (5 water, 0 fire)
- **Steam** (1 water, 3 fire)



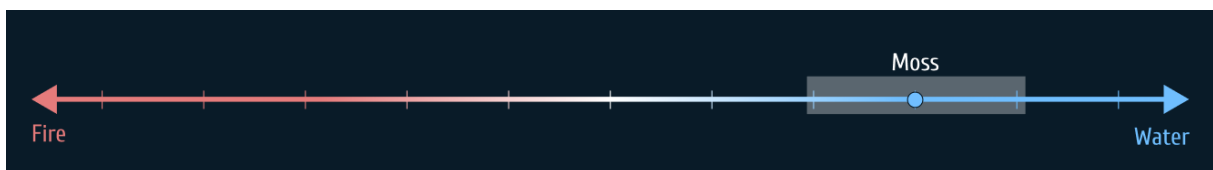
Initial state, mixture at 0



The player adds a **Water crystal**, the mixture moves 5 units right (5 water)



The player adds **Steam**, which moves the mixture $1 - 3 = -2$ units to the right (so 2 units left).



The mixture lands in the discovery zone, **Moss** is unlocked for further experimentation!

Player model

The player should notice the similarity between this one-dimensional elemental space and the number line. Both ends of this elemental space are color-coded for elements and evenly spaced notches indicate units. When the player sees an ingredient with composition (5 water, 0 fire), they should assume this moves the mixture 5 units to the right (the direction of water). Similarly, an ingredient with composition (0 water, 2 fire) should move the mixture 2 units to the left. When presented with an ingredient composed of both fire and water energy, they might reason like this:

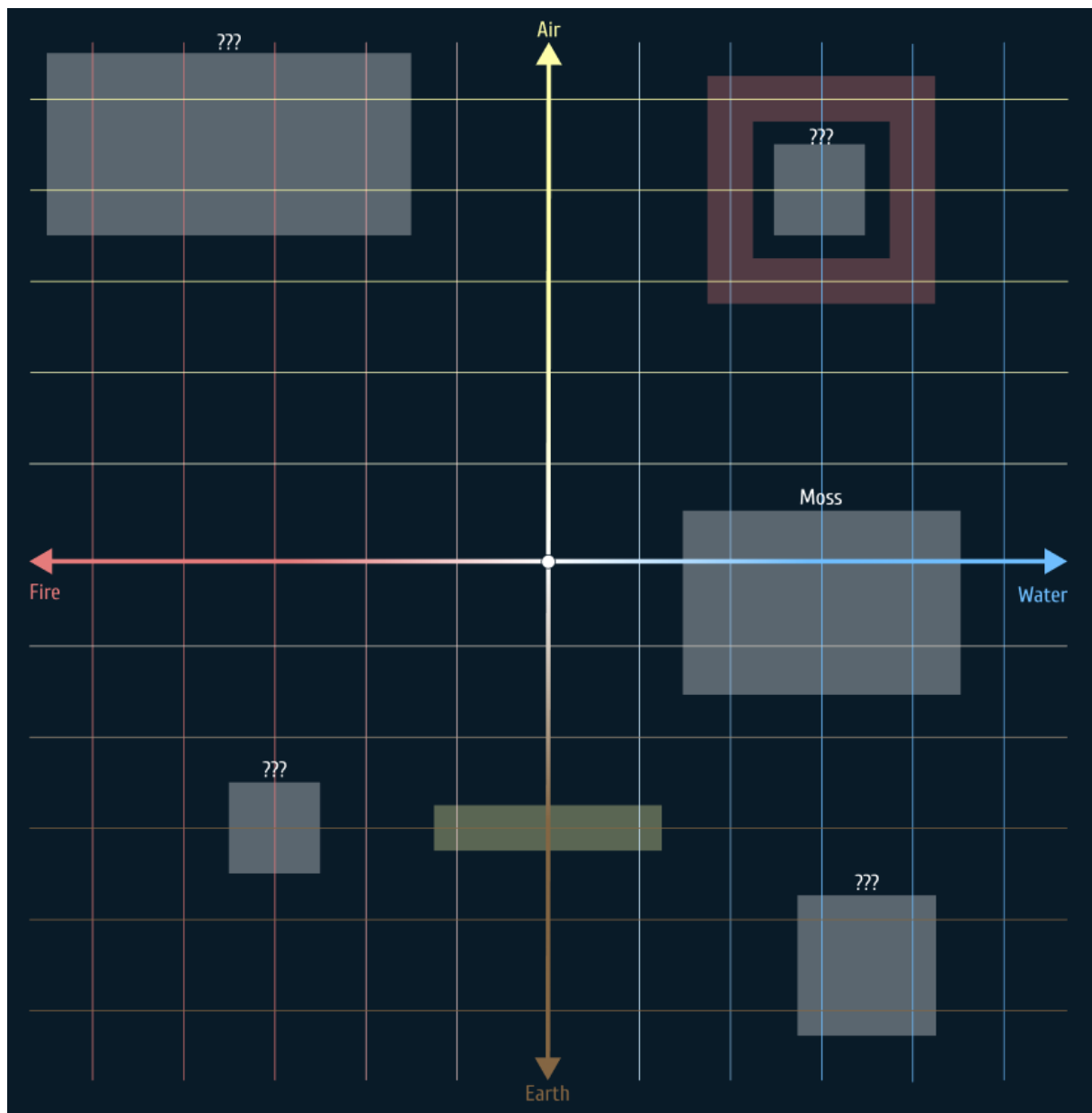
“Water ingredients move the mixture to the right, while fire ingredients move the mixture to the left. This new ingredient has both components, so they must interact in some way. Water can extinguish fire, so does water energy absorb fire energy? But fire can also turn water into steam, so maybe that’s the other way around? Or do they cancel each other out? If so, what’s the proportion? 1 water to 1 fire? 2 water to 1 fire?”.

Placing an ingredient composed of 1 water, 3 fire energy moves the mixture 2 units towards fire, which supports “cancelling out” hypothesis with ratio 1:1.

Air and Earth axes - Second dimension

When the player learns the basic principles of movement and elemental cancelling, the game adds another dimension – Air/Earth axis. The elemental space becomes a 2D map of alchemy. Ingredients can now have two elemental energies. For instance, an ingredient with 2 water, 3 earth energy moves the mixture 2 units right and 3 units down.

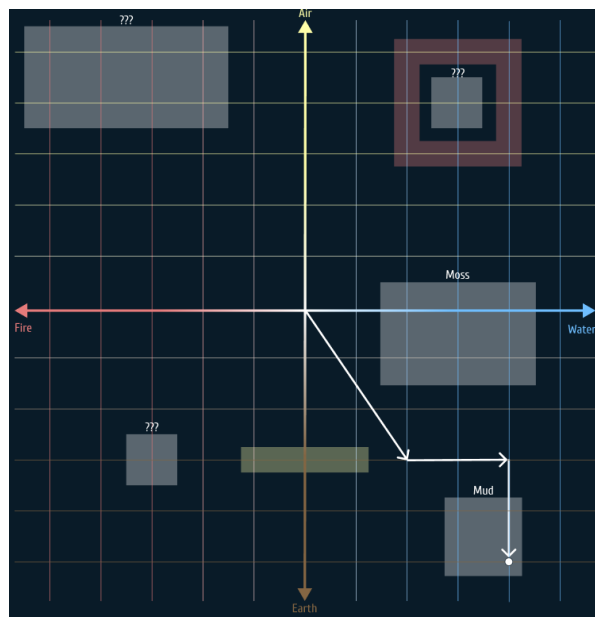
The discovery zones become rectangles, which increases the solution space, as more coordinate pairs correspond to the same result. Later in the game, colorful zones will also appear on the map. Look at the example alchemy map below:



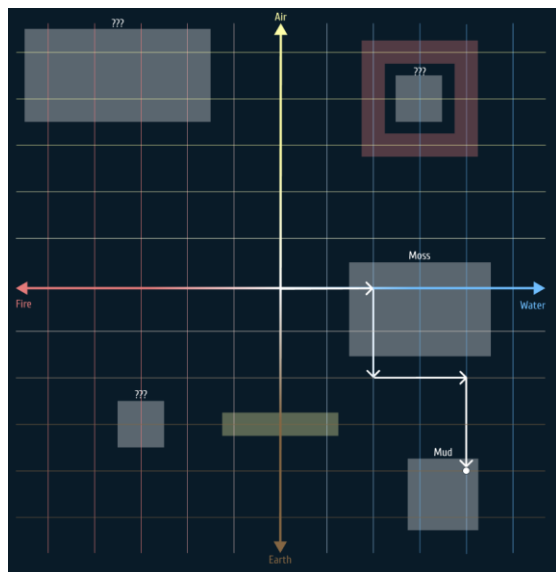
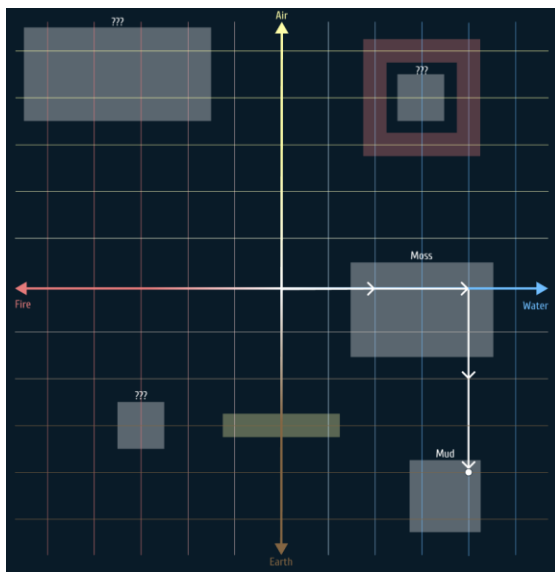
The experiment starts at the origin (0,0), where all elements are in balance. Player's goal is to arrive at any of the discovery zones using available ingredients. The ingredients can

be thought of as vectors in 2D space. An ingredient with composition (*air*, *earth*, *water*, *fire*) becomes a vector (*air* - *earth*, *water* - *fire*).

They might construct a path like this to discover **Mud**:



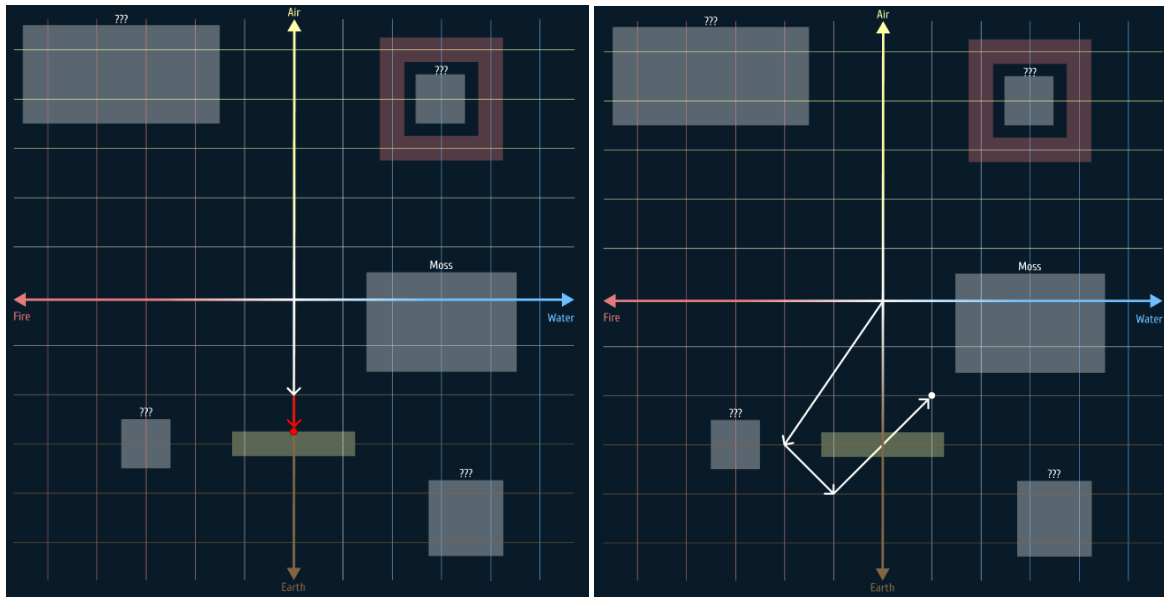
Notice that this is not the only solution: going twice right and twice down also is a valid path, albeit a longer one (4 vectors). See two examples of valid, longer paths below.



Elemental zones - First obstacles

Besides discovery zones, there are also colorful rectangles in 2D space. These are Elemental zones and can only be traversed in one direction – in the direction of their respective elements. For yellow zones, this means that the air component must be

stronger than the earth component. Compare these two paths, first invalid and the other valid, for demonstration.



The path on the left is invalid, because the second ingredient's earth component is stronger than the air component (its vector is pointing in downward direction). The path on the right is valid, since the third ingredient has more air than earth energy (its vector is pointing upwards). Bigger elemental zones create one-way corridors.

Player model

The player might reason like this:

“The game won't let me cross this zone with this ingredient. Should I just avoid those zones? The top-right red area surrounds a discovery zone, so there must be a way to get through these eventually. Let's see, this area is yellow. It might have something to do with air energy, since air is yellow. The red one should be somehow connected with fire. Is the yellow area blocking the ingredient because there is no air energy in this ingredient? Or is it pushing the vector back and I need a stronger earth ingredient? Or perhaps it's only passable in the up direction, since air is up? Or do I need to unlock another alchemy station that allows me to pass these areas?”

Aether zones – adding the Third Dimension

The fifth classical element is Aether. Ingredients cannot have Aether energy, instead the map contains special Aether zones, color-coded with purple. These zones allow the mixture to ascend and descent to another layer of the map. These should be thought of as teleporters rather than stairs, since those areas won't have a corresponding ascend/descent pair. When the mixture reaches Aether zone, it is moved to another layer

The image displays two diagrams, Layer 0 and Layer 1, illustrating the evolution of a neural network's internal representation across two layers. Both diagrams are set against a dark blue background with a light gray grid.

Layer 0: This diagram shows a 2D coordinate system with a horizontal axis labeled "Fire" (red arrow pointing left) and "Water" (blue arrow pointing right), and a vertical axis labeled "Air" (yellow arrow pointing up) and "Earth" (yellow arrow pointing down). Several colored rectangles represent different concepts: a gray rectangle labeled "Gunpowder" in the top-left; a purple rectangle labeled "Ascend" in the top-right, enclosed by a brown rectangle; a gray rectangle labeled "Moss" in the middle-right; a purple rectangle labeled "Descend" in the bottom-left; a green rectangle in the bottom-center; and a gray rectangle labeled "Mud" in the bottom-right.

Layer 1: This diagram shows a similar 2D coordinate system. The rectangles from Layer 0 are transformed or combined: "Gunpowder" is now a gray rectangle labeled "???" in the top-left; "Ascend" is a small gray rectangle labeled "???" in the middle-left; "Moss" is a gray rectangle labeled "???" in the top-center; "Descend" is a small gray rectangle labeled "???" in the middle-right; the green rectangle is now a large green L-shape labeled "???" in the bottom-center; and "Mud" is a gray rectangle labeled "???" in the bottom-center. A new blue vertical rectangle appears on the right side. A white dot is present in the top-right area.

By discovering stronger ingredients, the player can explore the alchemical map further away from the origin. However, they soon realize that using limited number of ingredients makes it very hard to navigate, with Discovery zones out of reach and blocked by Elemental zones.

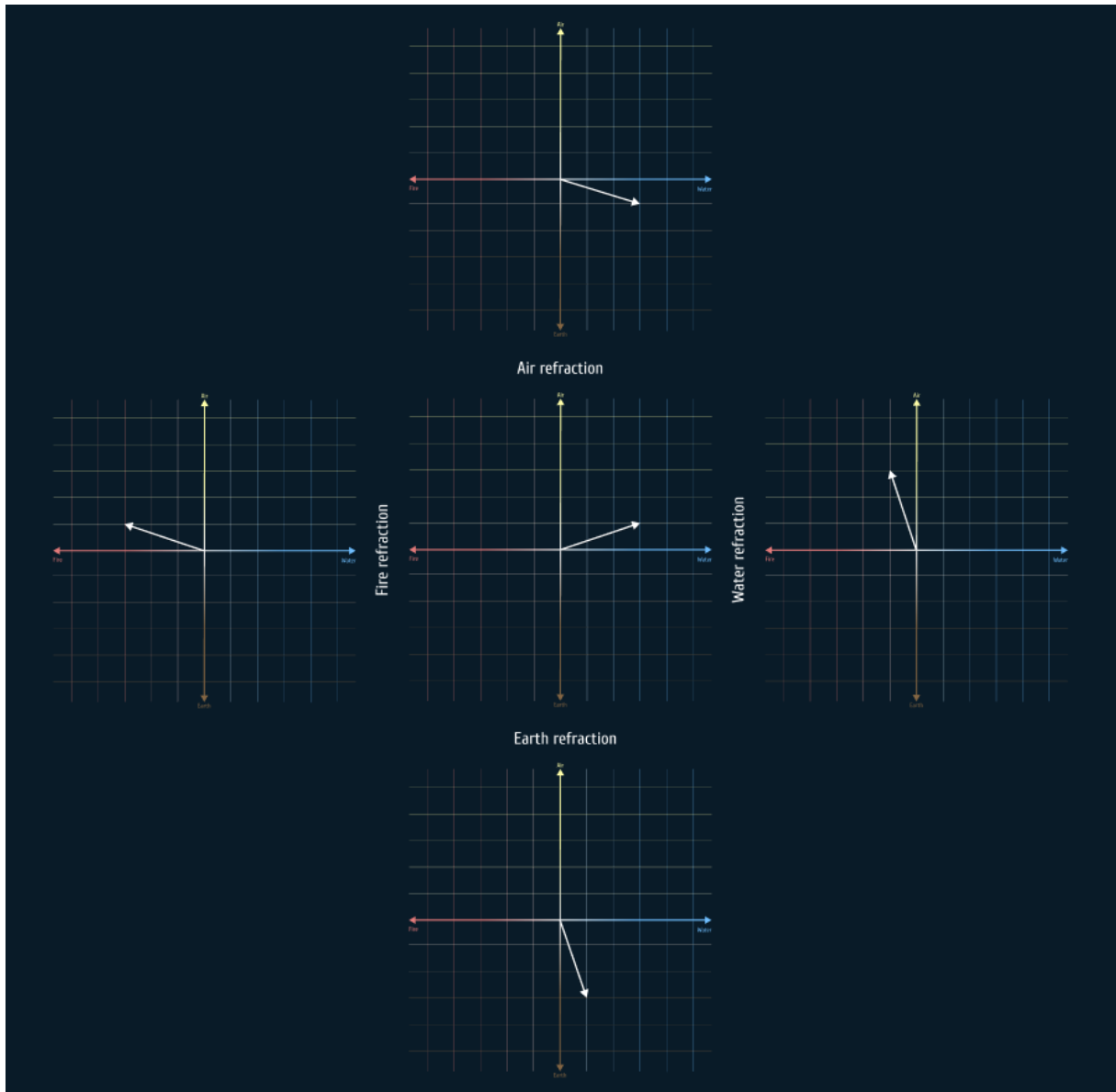
Note that each extra ingredient increases the problem space exponentially, so careful balancing is required regarding when better fuel becomes available.

Besides the cauldron, other alchemical stations are also available. These can be used to alter the energy of ingredients in various ways. Possible transformations include:

- Each processing station has 4 modes of working, associated with elemental energy.

For instance, pressing the ingredient using fire mode scales its vector by a factor of 5, using air scales by a factor of 4, water by a factor of 3, and earth by a factor of 2.

Refracting an ingredient in fire mode flips its vector horizontally, in air mode - vertically, in water mode rotates the vector 90 degrees counterclockwise, in earth mode rotates 90 degrees clockwise.



The starting vector is in the middle - (3, 1). Refracting with fire (left) transforms it into (-3, 1), with air (up) into (3, -1), with water (right) into (-1, 3), with earth (down) into (1, -3).

The transformations should be limited to one per ingredient to avoid exponential complexity growth.

Open questions

The core system establishes a foundation for knowledge-driven alchemy. The following questions represent key areas for further iteration and validation.

1. Learning and discovery:
 - a. How much of the elemental movement logic should player infer through experimentation versus learn through explicit tutorials?
 - b. At what point should navigating the elemental space feel like satisfying problem-solving rather than hypothesizing and discovering new rules?
2. Challenge and complexity:
 - a. How should ingredient capacity, discovery, elemental and Aether zones evolve to maintain meaningful problem solving without overwhelming the player?
3. Solution diversity:
 - a. As players unlock more ingredients and processing stations, how can we ensure that challenges continue to support multiple viable approaches?
 - b. How should discovery zones be designed to reward experimentation without encouraging brute-force ingredient combinations?
4. Failure and feedback:
 - a. What feedback can help players understand why a path failed and make a more informed next attempt, without simply revealing the answer?
 - b. How can we inform the player that some discovery zones are unreachable without unlocks from other systems?
 - c. Should we allow the player to undo/redo the last operation?
5. Spatial readability:
 - a. How can we represent 3D elemental space in a way that is easy to understand and navigate?
 - b. What visual and interaction conventions will help players reason about elemental zones and Aether transitions?
6. Additional features:
 - a. Should we allow the player to start new experiments on any previously visited layer?
 - b. Should we introduce a mechanic, where a vector splits into two separate paths? This would support one of the mechanic-driven endings.
7. Validation:
 - a. How can prototyping and playtesting reveal whether players genuinely understand elemental movement and cancellation, rather than relying on guesswork?

These questions define the next stage of development: validating the player's mental model, tuning the complexity curve, and ensuring that every discovery feels earned.